CE4011 Assignment 4

Special Topics in Civil Engineering: Structural Analysis Software Development

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Chapter 1

Revised Object-Oriented Structural Analysis Program

1.1 Introduction

The object-oriented structural analysis program developed in Assignment 3 is extended in this work to incorporate thermal loading effects and support settlements. The original solver, based on the Direct Stiffness Method (DSM), supported linear static analysis of two-dimensional frame and truss structures using a modular and extensible class-based architecture.

The present revision introduces two additional capabilities that are essential in practical structural analysis. Thermal actions are incorporated to account for temperature-induced deformation and stress, while support settlements are modeled as prescribed displacements at restrained degrees of freedom. These extensions are implemented with minimal disruption to the existing solver, preserving its original architecture and workflow.

The objective of this chapter is to document the design modifications, implementation strategy, and theoretical integration of these new features within the existing object-oriented framework.

1.2 Analysis and Design Decisions

1.2.1 Thermal vs Settlement Modeling

A key design decision was to treat thermal loading and support settlement as fundamentally different types of effects.

Thermal actions are internal to structural members and are therefore modeled at the element level using equivalent nodal loads derived from fixed-end forces. In contrast, support settlements represent imposed displacements at boundary conditions and must be incorporated at the global system level.

This distinction ensures that:

- Thermal effects are handled locally within element formulations
- Settlement effects are handled globally through system partitioning

1.2.2 Modification Strategy

The solver was extended using the following approach:

- The `Material` class was extended to include the coefficient of thermal expansion α
- The `Section` class was extended to include section depth d for thermal gradient calculations
- The `Node` class was extended to store prescribed displacements
- The `FrameElement` and `TrussElement` classes were extended to compute thermal fixed-end forces
- The `Structure` class was extended to solve the partitioned system for settlement

This approach preserved the original solver architecture while enabling the required functionality.

1.3 Revised UML and Class Modifications

1.3.1 Updated Class Responsibilities

The primary modifications to the class structure are summarized below:

- **Node:** Stores prescribed displacements for settlement modeling
- **Material:** Includes thermal expansion coefficient α
- **Section:** Includes section depth d
- **FrameElement:** Computes thermal axial forces and bending moments
- **TrussElement:** Computes thermal axial forces only
- **Structure:** Handles partitioned system solution and reaction computation

1.3.2 UML/Class Diagram

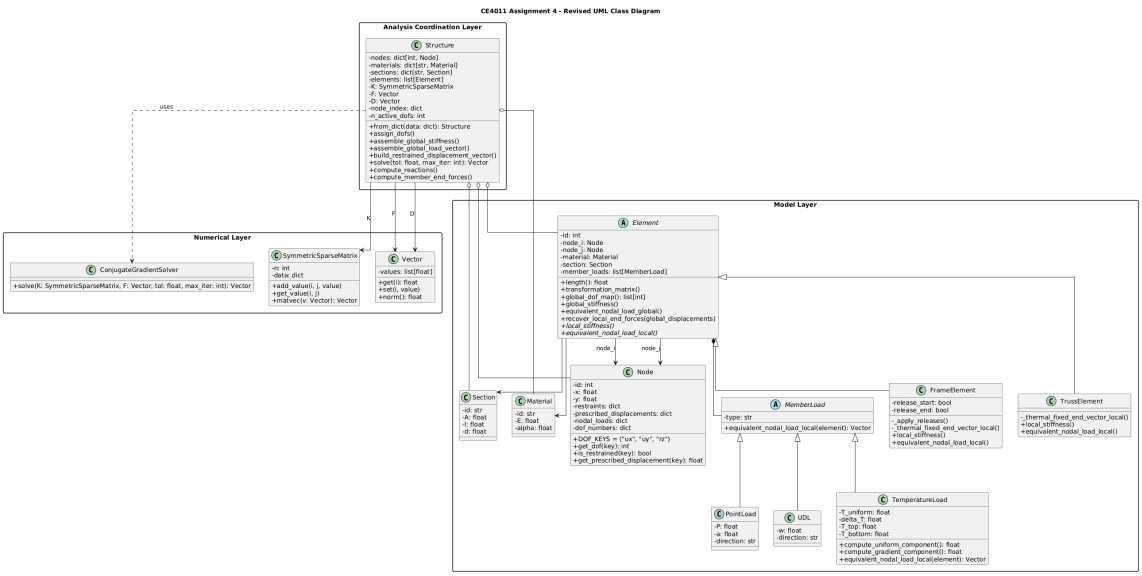

Figure 1.1: Revised UML/Class Diagram showing thermal and settlement extensions

1.4 Thermal Loading Extension

1.4.1 Modeling Approach

Thermal effects are incorporated using the fixed-end force method. Temperature-induced strains are first computed and then converted into equivalent nodal forces acting on the structure. These forces are assembled into the global load vector without modifying the stiffness matrix.

1.4.2 Uniform Temperature Change

A uniform temperature change produces an axial strain:

$$\epsilon_{th} = \alpha \Delta T \tag{1.1}$$

For a fully restrained member, this results in an axial force:

$$N_t = EA\alpha\Delta T \tag{1.2}$$

This force is applied as an equivalent nodal load in the axial direction.

1.4.3 Thermal Gradient Effects

A temperature gradient across the section depth induces curvature:

$$\kappa = \alpha \frac{\Delta T}{d} \tag{1.3}$$

The resulting bending moment is:

$$M_t = EI\alpha \frac{\Delta T}{d} \tag{1.4}$$

This moment is represented as equivalent nodal moments at the element ends.

1.4.4 Implementation in Elements

Thermal loads are implemented by computing fixed-end force vectors within each element and converting them into equivalent nodal loads. For frame elements, both axial forces and bending moments are considered, while for truss elements only axial effects are included.

1.5 Support Settlement Extension

1.5.1 Settlement Modeling Approach

Support settlements are modeled as prescribed displacements at restrained degrees of freedom. Unlike external loads, settlements modify the displacement boundary conditions and must be incorporated into the global system equations.

1.5.2 Partitioned System Formulation

The global stiffness equation is partitioned as:

$$\begin{bmatrix} K_{ff} & K_{fr} \\ K_{rf} & K_{rr} \end{bmatrix} \begin{Bmatrix} u_f \\ u_r \end{Bmatrix} = \begin{Bmatrix} F_f \\ F_r \end{Bmatrix} \tag{1.5}$$

For non-zero prescribed displacements, the system is solved as:

$$K_{ff} u_f = F_f - K_{fr} u_r \tag{1.6}$$

Support reactions are obtained from:

$$F_r = K_{rf} u_f + K_{rr} u_r \tag{1.7}$$

This formulation is implemented within the `Structure` class by constructing the restrained displacement vector and modifying the right-hand side of the system.

1.6 Updated Relationships and Workflow

1.6.1 Analysis Workflow

The revised analysis procedure consists of the following steps:

1. Assign degrees of freedom to all nodes
2. Assemble the global stiffness matrix
3. Assemble the global load vector including thermal loads
4. Construct the restrained displacement vector for settlement
5. Modify the right-hand side of the system
6. Solve for unknown displacements
7. Compute support reactions
8. Recover member-end forces

1.6.2 Workflow Illustration

Figure 1.2: Updated analysis workflow incorporating thermal and settlement effects

1.7 Input Format

1.7.1 Data Schema Overview

The solver accepts structured input defining nodes, materials, sections, elements, and loads. Thermal loads are specified as member loads, while settlements are defined as prescribed displacements at nodes.

1.7.2 Example Input Snippet

Listing 1.1: Example thermal and settlement input

```
{
  "elements": [
    {
      "type": "frame",
      "member_loads": [
        {"type": "thermal", "T_top": 50, "T_bottom": 0}
      ]
    }
  ],
  "prescribed_displacements": {
    "node_2": {"uy": -0.002}
  }
}
```

1.8 Output Format

1.8.1 Reported Quantities

The solver produces the following outputs:

- Nodal displacements
- Support reactions
- Member-end forces in local coordinates

1.8.2 Output Table Structure

Table 1.1: Output quantities

Quantity	Unit	Description
Displacement	m, rad	Nodal DOFs
Reaction	N, Nm	Support forces and moments
Member-end forces	N, Nm	Local element forces

1.9 GitHub Repository Summary

The complete implementation of the revised solver is available in the project repository.

The repository includes:

- Core solver implementation (Assignment 3 and extensions)
- Thermal and settlement modules
- Unit and integration tests
- Example input cases

Chapter 2

Testing and Verification

2.1 Testing Framework

2.2 Unit Tests

2.2.1 Unit Test 1: Thermal Load Component

2.2.2 Unit Test 2: Support Settlement Component

2.3 Interface and Integration Tests

2.3.1 Integration Test 1: Thermal Loading Pipeline

2.3.2 Integration Test 2: Settlement + Structural Solver Pipeline

2.4 Thermal Verification Cases

2.4.1 Case Definition and Assumptions

2.4.2 Verification Results

Table 2.1: Placeholder: Thermal verification results

Quantity	Program Result	Benchmark Result	Difference (%)
TODO	TODO	TODO	TODO

2.5 Settlement Verification Cases

2.5.1 Case Definition and Assumptions

2.5.2 Verification Results

Table 2.2: Placeholder: Settlement verification results

Quantity	Program Result	Benchmark Result	Difference (%)
TODO	TODO	TODO	TODO

2.6 Summary Table of Verification Results

Table 2.3: Placeholder: Consolidated verification summary

Test ID	Test Type	Target Feature	Status	Remarks
UT-1	Unit	Thermal loading	TODO	TODO
UT-2	Unit	Settlement modeling	TODO	TODO
IT-1	Integration	Thermal pipeline	TODO	TODO
IT-2	Integration	Settlement pipeline	TODO	TODO

Chapter 3

Analysis of Assignment Structures

3.1 Overview of Assignment Structures

3.2 Structure (a): Settlement Case

3.2.1 Model Description

3.2.2 Input Assumptions

3.2.3 Solver Results

Table 3.1: Placeholder: Structure (a) solver results

Output Type	Location	Value	Unit
Nodal displacement	TODO	TODO	TODO
Support reaction	TODO	TODO	TODO
Member-end force	TODO	TODO	TODO

3.2.4 Benchmark Comparison (SAP2000 / Hand Calculation)

Figure 3.1: Placeholder: SAP2000/benchmark output for Structure (a)

3.3 Structure (b): Thermal Case

3.3.1 Model Description

3.3.2 Input Assumptions

3.3.3 Solver Results

Table 3.2: Placeholder: Structure (b) solver results

Output Type	Location	Value	Unit
Nodal displacement	TODO	TODO	TODO
Support reaction	TODO	TODO	TODO
Member-end force	TODO	TODO	TODO

3.3.4 Benchmark Comparison (SAP2000 / Hand Calculation)

Figure 3.2: Placeholder: SAP2000/benchmark output for Structure (b)

3.4 Discussion of Discrepancies

3.4.1 Observed Differences

3.4.2 Potential Sources of Error

3.4.3 Engineering Interpretation

Chapter 4

Conclusion

4.1 Summary of Extensions

4.2 Solver Capability Improvements

4.3 Verification Outcome

4.4 Final Remarks

Chapter 5

References